

Lab 6: AI-Based Code Completion – Classes, Loops, and Conditionals

Week 3 – Wednesday

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Task 1: Classes – Student Class

Prompt Used:

"Generate a Python Student class with name, roll number, branch, and a method to display details."

```
class Student:
    def __init__(self, name, roll_no, branch):
        self.name = name
        self.roll_no = roll_no
        self.branch = branch

    def display_details(self):
        print("Name:", self.name)
        print("Roll Number:", self.roll_no)
        print("Branch:", self.branch)

s1 = Student("Eshwar", "2303A51495", "CSE")
s1.display_details()
```

Output:

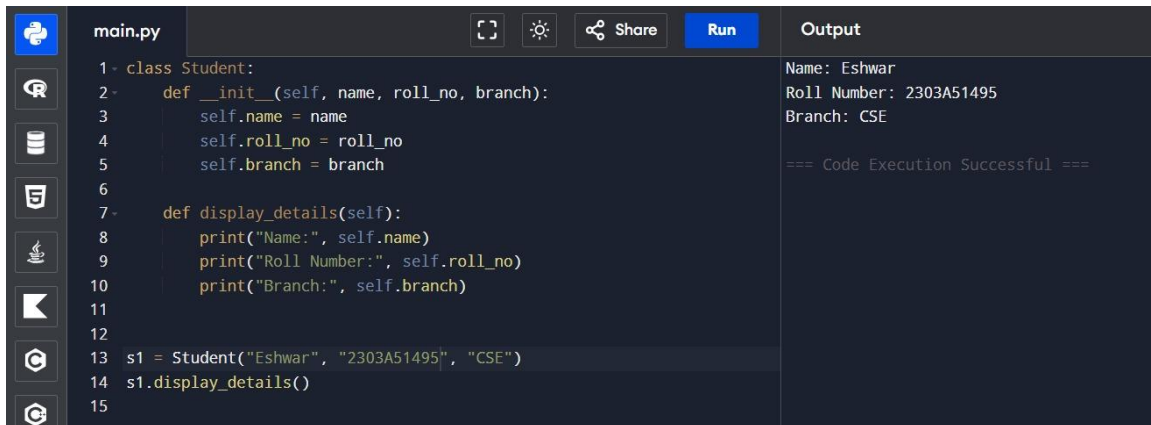
Name: Eshwar

Roll Number: 2303A51495

Branch: CSE

Analysis:

The AI-generated class is well-structured, readable, and correctly uses a constructor and instance method.



The screenshot shows a Python IDE with a file named 'main.py'. The code defines a 'Student' class with an '__init__' method that takes 'name', 'roll_no', and 'branch' as arguments and assigns them to instance variables. It also has a 'display_details' method that prints the values of these variables. The code is executed, and the output is shown in a separate pane. The output displays the name 'Eshwar', roll number '2303A51495', and branch 'CSE'. A message '=== Code Execution Successful ===' is also shown.

```
1 class Student:
2     def __init__(self, name, roll_no, branch):
3         self.name = name
4         self.roll_no = roll_no
5         self.branch = branch
6
7     def display_details(self):
8         print("Name:", self.name)
9         print("Roll Number:", self.roll_no)
10        print("Branch:", self.branch)
11
12
13 s1 = Student("Eshwar", "2303A51495", "CSE")
14 s1.display_details()
15
```

Output

```
Name: Eshwar
Roll Number: 2303A51495
Branch: CSE

=== Code Execution Successful ===
```

Task 2: Loops – Multiples of a Number

Prompt Used:

"Generate Python code to print first 10 multiples of a number using loops."

Using for loop:

```
def multiples_for(n):
    for i in range(1, 11):
        print(n * i)
```

multiples_for(5)

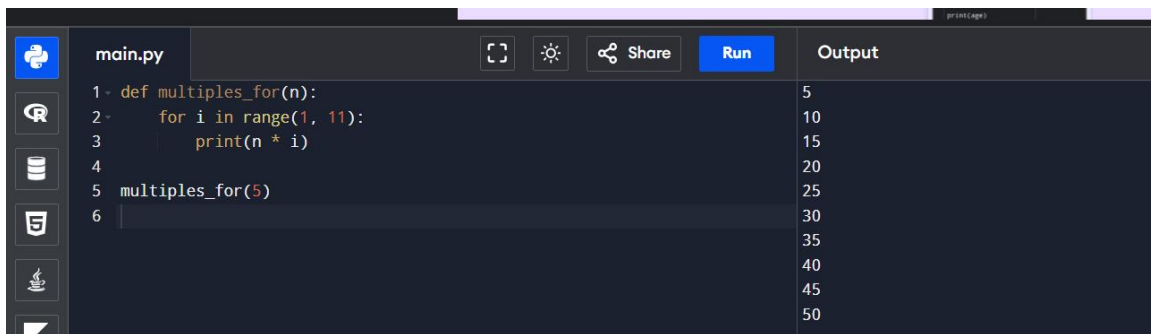
Using while loop:

```
def multiples_while(n):
    i = 1
    while i <= 10:
        print(n * i)
        i += 1
```

multiples_while(5)

Output:

5 10 15 20 25 30 35 40 45 50



The screenshot shows a code editor with a file named 'main.py'. The code defines a function 'multiples_for(n)' that uses a 'for' loop to print multiples of 'n' from 1 to 10. The function is called with 'multiples_for(5)'. The output on the right shows the numbers 5, 10, 15, 20, 25, 30, 35, 40, 45, and 50.

```
1 def multiples_for(n):
2     for i in range(1, 11):
3         print(n * i)
4
5 multiples_for(5)
6
```

Output:

5
10
15
20
25
30
35
40
45
50

Analysis:

The for loop is concise and readable, while the while loop provides explicit control over iteration.

Task 3: Conditional Statements – Age Classification

Prompt Used:

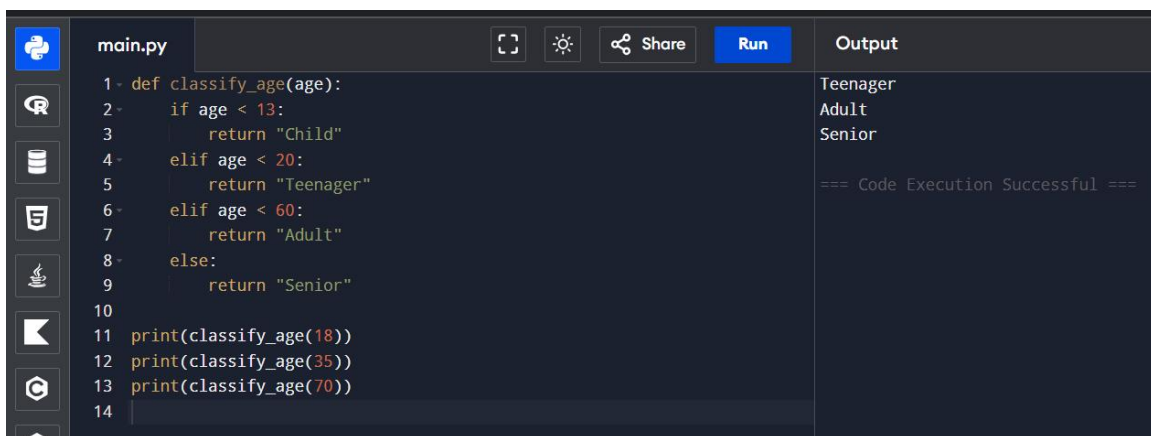
"Generate Python code to classify age using if-elif-else."

```
def classify_age(age):
    if age < 13:
        return "Child"
    elif age < 20:
        return "Teenager"
    elif age < 60:
        return "Adult"
    else:
        return "Senior"

print(classify_age(18))
```

Output:

Teenager



The screenshot shows a code editor with a file named 'main.py'. The code defines a function 'classify_age(age)' that uses 'if-elif-else' statements to classify an age into 'Child', 'Teenager', 'Adult', or 'Senior'. The function is called with 'classify_age(18)', 'classify_age(35)', and 'classify_age(70)'. The output on the right shows 'Teenager', 'Adult', and 'Senior' respectively, followed by a success message.

```
1 def classify_age(age):
2     if age < 13:
3         return "Child"
4     elif age < 20:
5         return "Teenager"
6     elif age < 60:
7         return "Adult"
8     else:
9         return "Senior"
10
11 print(classify_age(18))
12 print(classify_age(35))
13 print(classify_age(70))
14
```

Output:

Teenager
Adult
Senior

=== Code Execution Successful ===

Explanation:

The conditions are checked in sequence. The first matching condition determines the age group.

Task 4: For and While Loops – Sum of First n Numbers

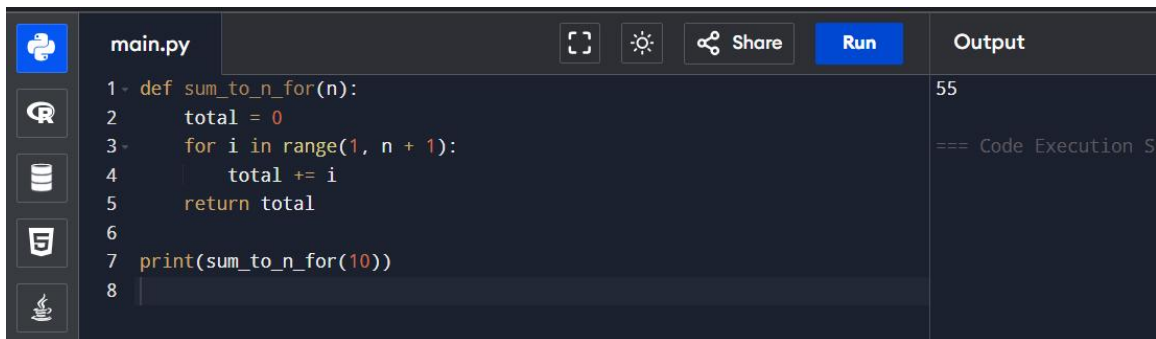
Prompt Used:

"Generate Python code to find sum of first n natural numbers using loops."

Using for loop:

```
def sum_to_n_for(n):  
    total = 0  
    for i in range(1, n + 1):  
        total += i  
    return total
```

```
print(sum_to_n_for(10))
```



The screenshot shows a Python IDE interface. On the left, there is a sidebar with icons for Python, Jupyter, and other tools. The main area displays a file named 'main.py' with the following code:

```
1 def sum_to_n_for(n):  
2     total = 0  
3     for i in range(1, n + 1):  
4         total += i  
5     return total  
6  
7 print(sum_to_n_for(10))  
8
```

At the top right of the code editor, there are buttons for 'Run' and 'Share'. To the right of the code editor, there is an 'Output' panel showing the result of the execution:

```
55  
  
=== Code Execution S
```

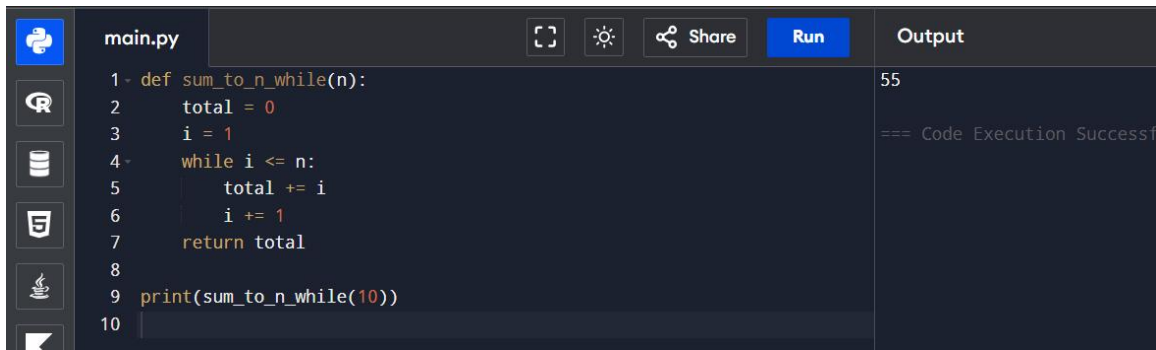
Using while loop:

```
def sum_to_n_while(n):  
    total = 0  
    i = 1  
    while i <= n:  
        total += i  
        i += 1  
    return total
```

```
print(sum_to_n_while(10))
```

Output:

55

A screenshot of a code editor interface. The editor has a dark theme. On the left, there's a sidebar with icons for a file explorer, a search bar, and other tools. The main area shows a file named 'main.py' with the following Python code:

```
1 def sum_to_n_while(n):
2     total = 0
3     i = 1
4     while i <= n:
5         total += i
6         i += 1
7     return total
8
9 print(sum_to_n_while(10))
10
```

At the top right of the editor, there are buttons for 'Run', 'Share', and a settings icon. To the right of the code editor, there's an 'Output' panel showing the result of the execution:

```
55
=== Code Execution Successful ===
```

Analysis:

Both approaches are correct. Loop-based methods are simple, while mathematical formulas can be more efficient.

Task 5: Classes – Bank Account Class

Prompt Used:

"Generate a Python BankAccount class with deposit, withdraw, and check balance methods."

class BankAccount:

```
    def __init__(self, balance=0):
        self.balance = balance
```

```
    def deposit(self, amount):
        self.balance += amount
        print("Deposited:", amount)
```

```
    def withdraw(self, amount):
        if amount <= self.balance:
            self.balance -= amount
            print("Withdrawn:", amount)
        else:
            print("Insufficient balance")
```

```
    def check_balance(self):
        print("Current Balance:", self.balance)
```

```
account = BankAccount(1000)
account.deposit(500)
account.withdraw(300)
account.check_balance()
```

Output:

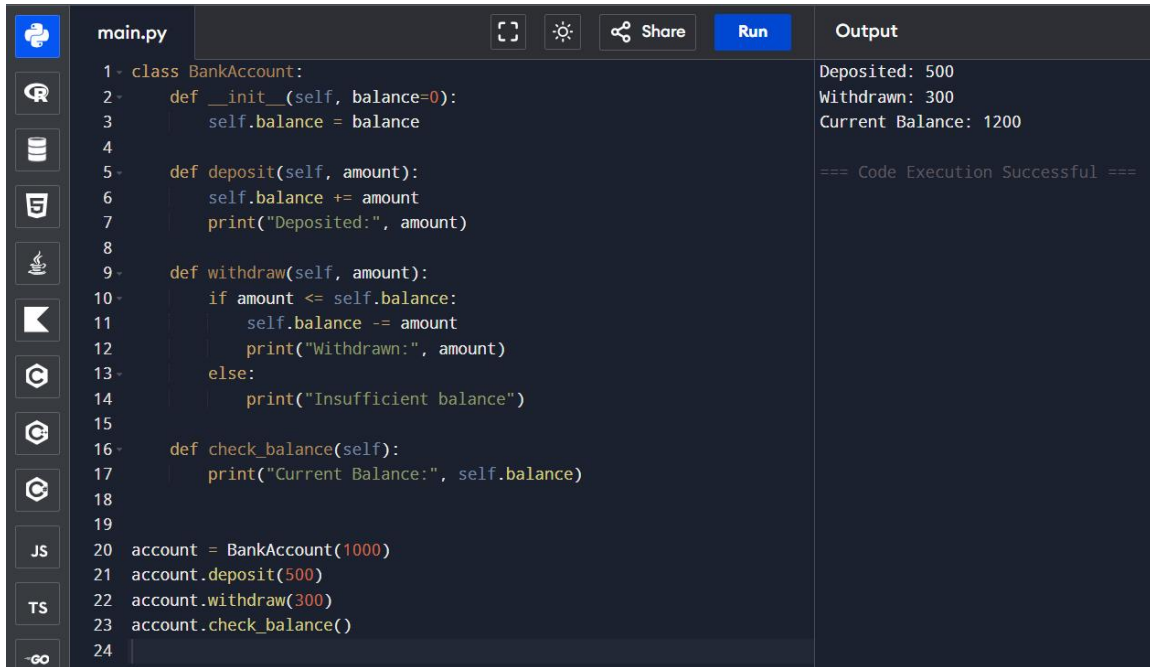
Deposited: 500

Withdrawn: 300

Current Balance: 1200

Explanation:

The class maintains account balance and updates it through deposit and withdraw methods.



The screenshot displays a code editor interface with a file named 'main.py'. The code defines a 'BankAccount' class with methods for deposit, withdraw, and check_balance. The main program creates an account with an initial balance of 1000, deposits 500, withdraws 300, and checks the balance. The output on the right shows the results of these operations: 'Deposited: 500', 'Withdrawn: 300', and 'Current Balance: 1200', followed by a success message.

```
1 class BankAccount:
2     def __init__(self, balance=0):
3         self.balance = balance
4
5     def deposit(self, amount):
6         self.balance += amount
7         print("Deposited:", amount)
8
9     def withdraw(self, amount):
10        if amount <= self.balance:
11            self.balance -= amount
12            print("Withdrawn:", amount)
13        else:
14            print("Insufficient balance")
15
16    def check_balance(self):
17        print("Current Balance:", self.balance)
18
19
20 account = BankAccount(1000)
21 account.deposit(500)
22 account.withdraw(300)
23 account.check_balance()
24
```

Output

Deposited: 500
Withdrawn: 300
Current Balance: 1200

=== Code Execution Successful ===

Overall Conclusion

This lab demonstrates how AI-assisted code completion helps in generating structured, readable, and correct Python programs. Human review is essential to ensure correctness and efficiency.